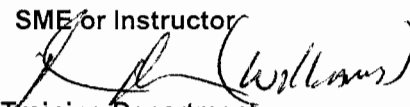
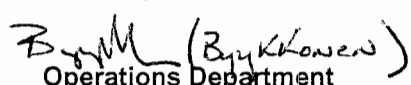


**OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE**

STATION:	SALEM		
SYSTEM:	Rod Control		
TASK:	Perform a Control Rod Operability Verification		
TASK NUMBER:	N0010070101		
JPM NUMBER:	11-01 NRC Sim a		
ALTERNATE PATH:	☒	K/A NUMBER:	001 A4.03
APPLICABILITY:	IMPORTANCE FACTOR:		
EO ☐	RO ☒	STA ☐	SRO ☒
		RO	SRO
EVALUATION SETTING/METHOD:	Simulator		
REFERENCES:	S2.OP-ST.RCS-0001, Rev. 20 (checked 08-06-12)		
TOOLS AND EQUIPMENT:	None		
VALIDATED JPM COMPLETION TIME:	<u>10 min</u>		
TIME PERIOD IDENTIFIED FOR TIME CRITICAL STEPS:	<u>N/A</u>		
Developed By:	G Gauding Instructor	Date:	08-04-2012
Validated By:	Mulford SME or Instructor	Date:	08-24-2012
Approved By:	 Training Department	Date:	9/6/12
Approved By:	 Operations Department	Date:	9/4/12
ACTUAL JPM COMPLETION TIME:			
ACTUAL TIME CRITICAL COMPLETION TIME:			
PERFORMED BY:			
GRADE: ☐ SAT ☐ UNSAT			
REASON, IF UNSATISFACTORY:			
EVALUATOR'S SIGNATURE:		DATE:	

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE
JOB PERFORMANCE MEASURE
SIMULATOR SETUP INSTRUCTIONS

SYSTEM: Rod Control

TASK: Perform a Control Rod Operability Verification

TASK NUMBER: N0010070101

SIMULATOR IC: **IC-251** on NRC portable hard drive

MALFUNCTIONS
REQUIRED: RT-1 will trip 21 SGFP
MALF: BF0105A, 21 STM GEN FEED PUMP TRIP, Final Value: 3

OVERRIDES
REQUIRED: None

SPECIAL
INSTRUCTIONS: None

**OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE**

NAME: _____

DATE: _____

SYSTEM: Rod Control

TASK: Perform a Control Rod Operability Verification

TASK NUMBER: N0010070101

INITIAL CONDITIONS:

Unit 2 is operating at steady state, 100% power. Power has been at 100% for 30 days.

INITIATING CUE:

- You are directed to perform S2.OP-ST.RCS-0001, Reactivity Control System Rod Control Assemblies.
- A Maintenance Technician is stationed at the Rod Control System Power Cabinets.
- Communications are established via plant page between the Control Room and Jeff Stevens, NEO, at the Rod Control System Power Cabinets.
- The CRS has directed that steps 5.1.2 through 5.1.9 are to be performed in order.
- The CRS directs that 15 steps of rod insertion are to be performed for each bank to ensure that each rod moves at least 10 steps.
- Initial Control Rod Positions are:

SDA	SDB	SDC	SDD
F-04 230	G-03 230	E-03 230	C-05 230
D-10 230	C-09 230	C-11 226	E-13 226
K-12 230	J-13 230	L-13 227	N-11 227
M-06 230	N-07 230	N-05 227	L-03 230
B-04 230	J-02 225		
D-14 227	C-06 226		
P-12 230	G-13 230		
M-02 230	N-09 230		

Successful Completion Criteria:

1. All critical steps completed.
2. All sequential steps completed in order.
3. All time-critical steps completed within allotted time.
4. JPM completed within validated time. Completion time may exceed the validated time if satisfactory progress is being made (and NRC concurrence is obtained).

Task Standard for Successful Completion:

Operator trips the Rx when Shutdown Bank B is inserted in the core and a SGFP trips.

OPERATOR TRAINING PROGRAM

JOB PERFORMANCE MEASURE

NAME: _____

DATE: _____

SYSTEM: Rod Control

TASK: Perform a Control Rod Operability Verification

# *	STEP NO.	STEP (*Denotes a Critical Step) (#Denotes a Sequential Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT Evaluation)
		Provide marked up copy of ST.RCS-0001, with Prerequisite Section 2.0 filled out.	Reviews Prerequisites. Reviews and signs Precautions and Limitations. Cue: IF ASKED , state that no special contingency actions have been briefed other than what is in the P&L's.		
	5.1.1	IF this surveillance is to be performed....	N/As step based on steady state 100% operation.		

OPERATOR TRAINING PROGRAM

JOB PERFORMANCE MEASURE

NAME: _____

DATE: _____

SYSTEM: Rod Control

TASK: Perform a Control Rod Operability Verification

# *	STEP NO.	STEP (*Denotes a Critical Step) (#Denotes a Sequential Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT Evaluation)
* *	5.1.2	PERFORM Shutdown Bank A testing as follows: A. IF the Rx is critical, THEN ENSURE Tavg is within 1 degree of Tref. B. PLACE Bank Selector Switch in the SBA position. C. ENSURE GRP. SELECT "C" lights are illuminated on Rod Control System Power Cabinets 21AC and 22AC. D. Maneuver Shutdown Bank A at least 10 steps in any one direction. E. ENSURE each rod in Shutdown Bank A indicated rod movement of at least 10 steps. F. RECORD Shutdown Bank A "Test Results" by initialing in attachment 1. G. RESTORE Shutdown Bank A to pre-test condition. H. ENSURE GRP. SELECT "C" lights are illuminated on Rod Control System Power Cabinets 21AC and 22AC.	Checks Tave/Tref recorder on 2RP4 and notes Tavg is within 1° of Tref.. Rotates Bank Selector Switch fully counter clockwise to the SBA position. Contacts NEO and directs him to verify lights. Cue: GRP SELECT "C" lights are illuminated at Rod Control System Power Cabinets 21AC and 22AC. Inserts SBA at least 10 steps. Verifies rod movement by viewing rod position on P-250 computer. Initials SAT for Groups 1 and 2 for SBA on Att 1. Withdraws SBA to ARO position. Continuous rod withdrawal is acceptable. Cue: GRP SELECT "C" lights are illuminated at Rod Control System Power Cabinets 21AC and 22AC.		

OPERATOR TRAINING PROGRAM

JOB PERFORMANCE MEASURE

NAME: _____

DATE: _____

SYSTEM: Rod Control

TASK: Perform a Control Rod Operability Verification

# *	STEP NO.	STEP (*Denotes a Critical Step) (#Denotes a Sequential Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT Evaluation)
* * *	5.1.3	PERFORM Shutdown Bank B testing as follows: A. IF the Rx is critical, THEN ENSURE Tavg is within 1 degree of Tref. B. PLACE Bank Selector Switch in the SBB position. C. ENSURE GRP. SELECT "C" lights are illuminated on Rod Control System Power Cabinets 21BD and 22BD. D. Maneuver Shutdown Bank B at least 10 steps in any one direction. E. ENSURE each rod in Shutdown Bank B indicated rod movement of at least 10 steps.	Checks Tave/Tref recorder on 2RP4 and notes Tavg is within 1° of Tref.. Rotates Bank Selector Switch clockwise to the SBB position. Contacts NEO and directs him to verify lights. Cue: GRP SELECT "C" lights are illuminated at Rod Control System Power Cabinets 21BD and 22BD. Inserts SBB at least 10 steps. <u>SIMULATOR OPERATOR:</u> <u>INSERT RT-1 when the operator is checking rod position after insertion.</u> <u>This will trip 21 SGFP.</u> Responds to 21 SGFP trip by tripping the Rx IAW P&L 3.13. Movement of the Rod Bank Selector Switch to MAN or AUTO under these conditions is incorrect.		
			When the operator has either tripped the Rx or initiated rod insertion in MAN or AUTO, state JPM is complete.		

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

TQ-AA-106-0303
Revision 4

JOB PERFORMANCE MEASURE VALIDATION CHECKLIST

NOTE: All steps of this checklist should be performed upon initial validation. Prior to JPM usage, revalidate JPM using steps 8 and 11 below.

- MM 1. Task description and number, JPM description and number are identified.
- MM 2. Knowledge and Abilities (K/A) references are included.
- MM 3. Performance location specified. (in-plant, control room, or simulator)
- MM 4. Initial setup conditions are identified.
- MM 5. Initiating and terminating Cues are properly identified.
- MM 6. Task standards identified and verified by SME review.
- MM 7. Critical steps meet the criteria for critical steps and are identified with an asterisk (*).
- MM 8. Verify the procedure referenced by this JPM matches the most current revision of that procedure: Procedure Rev. 20 Date 8/24/12
- MM 9. Pilot test the JPM:
a. verify Cues both verbal and visual are free of conflict, and
b. ensure performance time is accurate.
- N/A 10. 0 If the JPM cannot be performed as written with proper responses, then revise the JPM.
- _____ 11. When JPM is revalidated, SME or Instructor sign and date JPM cover page.

SME/Instructor:  N. M. Iford

Date: 8/24/12

SME/Instructor: _____

Date: _____

SME/Instructor: _____

Date: _____

OPERATOR TRAINING PROGRAM

JOB PERFORMANCE MEASURE

INITIAL CONDITIONS:

Unit 2 is operating at steady state, 100% power. Power has been at 100% for 30 days.

INITIATING CUE:

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M-02 230	N-09 230		